

ESPORTS MATCH OUTCOME PREDICTION USING DEEP LEARNING
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

Name of Title: ESPORTS MATCH OUTCOME PREDICTION USING DEEP LEARNING

1. Abstract:

Esports has emerged as a rapidly growing competitive industry, generating large volumes of data from player performance, team statistics, match history, and in-game events. Predicting the outcome of esports matches is challenging due to the dynamic nature of gameplay and the influence of multiple performance factors. This project presents a **deep learning-based approach for esports match outcome prediction**, using historical match data and relevant player and team statistics as input features. The proposed system applies data preprocessing, feature selection, and deep neural network techniques to identify patterns associated with winning and losing outcomes. The model is trained and evaluated using historical esports match datasets, with performance measured through accuracy, precision, recall, F1-score, and other classification metrics. By learning complex relationships within the data, the proposed approach aims to provide more reliable match predictions than traditional statistical methods. The system can assist players, analysts, coaches, and esports enthusiasts in understanding performance trends and estimating the likely outcome of future matches. Overall, this study demonstrates the potential of deep learning for data-driven decision-making and predictive analytics in the rapidly evolving esports domain.

2. INTRODUCTION:

Introduction

Esports, or electronic sports, has developed into a major competitive and entertainment industry, attracting millions of players and viewers worldwide. Professional esports competitions involve highly skilled teams and players who compete in games such as multiplayer online battle arenas (MOBAs), first-person shooters (FPS), and real-time strategy games. The outcome of an esports match can depend on several factors, including

previous match performance, player statistics, team coordination, game strategies, and individual player skills. Because of these complex and dynamic factors, accurately predicting match outcomes is a challenging task. The rapid growth of esports has also resulted in the availability of large amounts of historical match data. This data can contain information such as team win rates, player performance, rankings, match history, game statistics, and other relevant features. Machine learning techniques can analyze these datasets and identify patterns that may not be easily recognized through traditional analysis. Among these techniques, deep learning is particularly useful because it can learn complex relationships and patterns from large datasets with minimal manual intervention.

The project "Esports Match Outcome Prediction Using Deep Learning" focuses on developing a predictive model capable of estimating the outcome of esports matches based on historical and performance-related data. The collected data is first cleaned and preprocessed to remove inconsistencies and convert relevant information into suitable numerical features. A deep learning model is then trained using these features to classify matches according to their expected outcomes, such as win or loss.

The performance of the proposed model can be evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The results can help determine how effectively deep learning can be applied to esports prediction. Such a system can be useful for esports analysts, coaches, teams, researchers, and fans by providing data-driven insights into team performance and potential match outcomes.

Overall, this project demonstrates how artificial intelligence and deep learning can be applied to esports analytics. By utilizing historical data and advanced predictive techniques, the proposed system aims to improve the understanding of factors influencing esports match outcomes and provide a reliable framework for future research in sports and gaming analytics.

3. SOFTWARE REQUIREMENTS:

Software	Requirement
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.9 or above
IDE/Development Environment	Jupyter Notebook, Google Colab, or Visual Studio Code
Deep Learning Framework	TensorFlow/Keras or PyTorch
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Database/Storage	CSV/Excel files or MySQL, if required
Web Browser	Google Chrome, Microsoft Edge, or Firefox
Version Control	Git/GitHub, optional
Internet Connection	Required for dataset collection, Google Colab, and online resources

Signature of the Guide

Course Instructor